

TOCOL + UNIQUE in Excel

1. What is TOCOL + UNIQUE?

TOCOL and UNIQUE are two functions used together. TOCOL flattens a range with many rows and columns into a single column, and UNIQUE then removes the repeated values. Together they solve the problem of finding every different value that is spread across several columns, such as skills, tags, or product names. A Data Analyst uses them to build a clean list of distinct items from a grid of data in one formula.

2. Syntax

`=UNIQUE(TOCOL(array, [ignore], [scan_by_column]))`

Argument	Required?	What it accepts
array	Required	The range to flatten and check for repeats (for example A2:C5)
ignore	Optional	What TOCOL skips: 0 keep all (default), 1 ignore blanks, 2 ignore errors, 3 ignore blanks and errors
scan_by_column	Optional	FALSE (default) reads row by row. TRUE reads column by column. It only changes the order of the results.

Excel works from the inside out. TOCOL runs first and turns the range into one column. UNIQUE then keeps each value once, in the order it first appears.

3. Simple Example

Get the different fruits from a 3 × 3 grid:

	A	B	C	D	E
1	Apple	Banana	Apple		Apple
2	Orange	Grape	Banana		Banana
3	Mango	Orange	Kiwi		Orange
4					Grape
5					Mango
6					Kiwi

Formula in E1 only: `=UNIQUE(TOCOL(A1:C3, 1))`

TOCOL reads the 9 cells row by row into one column. UNIQUE removes the repeats of Apple, Banana, and Orange. **Result: 6 different fruits**

4. Another Example

List every skill in a grid that has blank cells:

	A	B	C	D	E
1	Skill 1	Skill 2	Skill 3		Unique Skills
2	Excel	SQL			Excel
3	SQL	Python	Excel		SQL
4	Excel				Python
5	Power BI	SQL	Python		Power BI

Formula in E2 only: `=UNIQUE(TOCOL(A2:C5, 1))` (the heading in E1 is typed)

The 1 in TOCOL skips the blank cells. Without it, the blanks would appear in the result as 0. **Result: Excel, SQL, Python, Power BI**

5. When to Use It

- Get a list of different items that are spread across several columns
- Build a clean list of skills, tags, or categories from a grid
- Create a list for a dropdown or a summary table
- Count how many different items exist across multiple columns
- Check which values appear anywhere in a multi-column range

6. Important Things to Remember

- Read the formula from the inside out: TOCOL flattens first, then UNIQUE removes repeats.
- By default, TOCOL keeps blank cells as 0, so a 0 can appear in the result. Use 1 to ignore blanks.
- Results appear in the order each value first appears. `scan_by_column` changes the reading direction and so the order.
- UNIQUE is not case-sensitive, but extra spaces make values different, so "Excel " and "Excel" count as separate values.
- The result spills into the cells below. They must be empty, or Excel shows #SPILL!.
- TOCOL is available in Microsoft 365 and Excel 2024, so this combination does not work in Excel 2021 or earlier.

7. Common Mistakes

Mistake: Leaving out the ignore value, so a 0 appears in the list for every blank cell. **Fix:** Use `TOCOL(A2:C5, 1)`.

Mistake: Flattening more columns than needed, such as TOCOL around a FILTER result with several columns. Values from every column appear in the list. **Fix:** Filter only the column you need: `=UNIQUE(FILTER(A2:A7, B2:B7="Electronics"))`.

Mistake: Putting the TOCOL settings in the wrong brackets, such as `=UNIQUE(TOCOL(A2:C5), 1)`. **Fix:** Keep them inside TOCOL: `=UNIQUE(TOCOL(A2:C5, 1))`.

Mistake: Seeing repeats in the result, such as "SQL" and "SQL ". **Fix:** Hidden spaces make them different. Clean the data with TRIM.

8. Data Analyst Tip

Add other functions around the formula. `=SORT(UNIQUE(TOCOL(A2:C5, 1)))` gives the list in alphabetical order, and `=COUNTA(UNIQUE(TOCOL(A2:C5, 1)))` counts the different items, which is 4 in the second example.